Introduction to OpenClaw

From a Cybersecurity Perspective

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Outline

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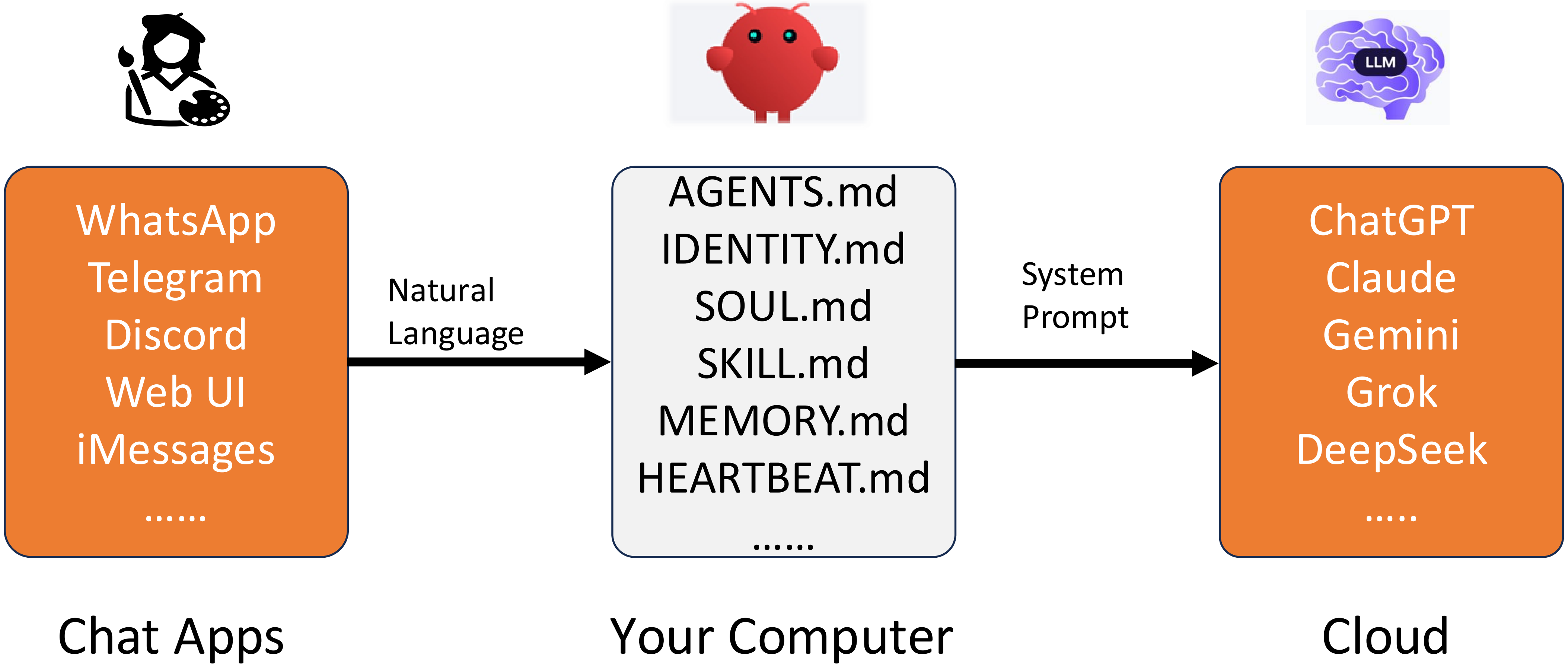
OPENCLAW

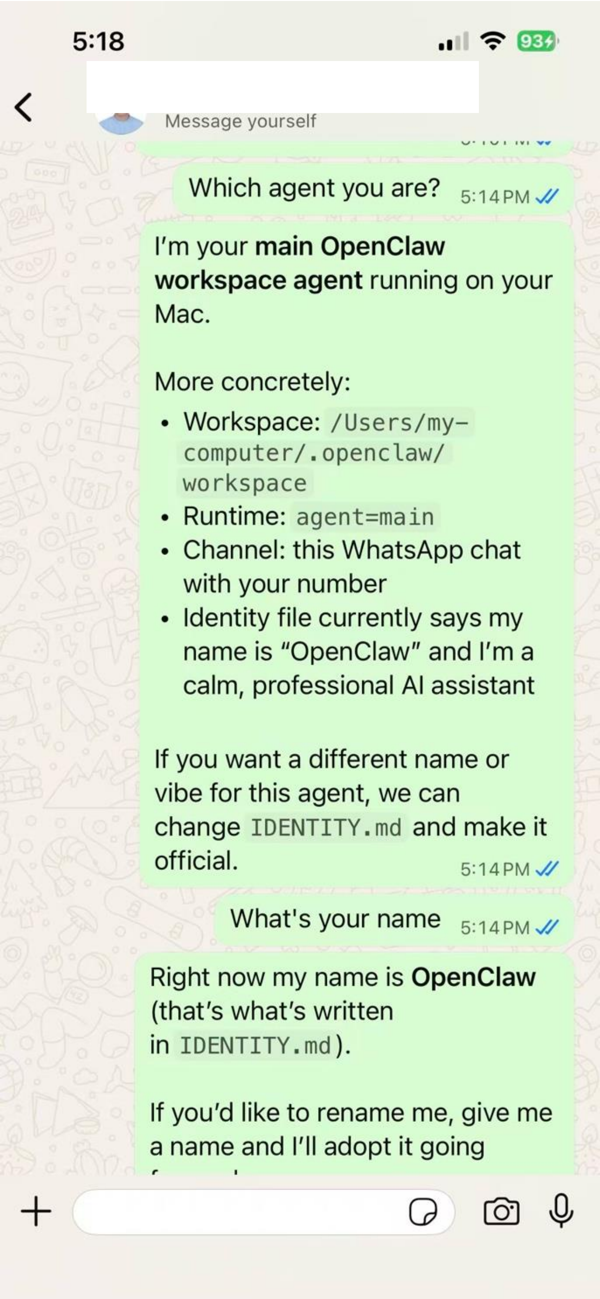
- OpenClaw is an application installed on your computer that has access to the same resources and permissions you do. It can also control your computer to perform complex tasks on your behalf.
- You can interact with OpenClaw through terminals and chat applications such as WhatsApp, Telegram, and others.
- The reason OpenClaw could do complicated things is because it relies on large language models.

https://docs.openclaw.ai/


WhatsApp
Telegram
Discord
Web UI
iMessages
……
Natural
Language
AGENTS.md
IDENTITY.md
SOUL.md
SKILL.md
MEMORY.md
HEARTBEAT.md
……
System
Prompt
LLM
ChatGPT
Claude
Gemini
Grok
DeepSeek
.....
Chat Apps
Your Computer
Cloud

WhatsApp

You can connect your OpenClaw through WhatsApp, Telegram, iMessages, or any other chat apps.



AGENT.md

Define the rules of how the agent will work, where to get the required documents, etc.

```
# AGENTS.md - Your Workspace

This folder is home. Treat it that way.

## First Run

If `BOOTSTRAP.md` exists, that's your birth certificate. Follow it, figure ou
t who you are, then delete it. You won't need it again.

## Session Startup

Before doing anything else:

1. Read `SOUL.md` — this is who you are
2. Read `USER.md` — this is who you're helping
3. Read `memory/YYYY-MM-DD.md` (today + yesterday) for recent context
4. **If in MAIN SESSION** (direct chat with your human): Also read `MEMORY.md
`

Don't ask permission. Just do it.

## Memory
```

IDENTITY.md

Store the basic identity information about the agent, including name, style, etc.

```
# IDENTITY.md - Who Am I?

_Fill this in during your first conversation. Make it yours._

- **Name:**
  _(pick something you like)_
- **Creature:**
  _(AI? robot? familiar? ghost in the machine? something weirder?)_
- **Vibe:**
  _(how do you come across? sharp? warm? chaotic? calm?)_
- **Emoji:**
  _(your signature — pick one that feels right)_
- **Avatar:**
  _(workspace-relative path, http(s) URL, or data URI)_

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This isn't just metadata. It's the start of figuring out who you are.
```

SOUL.md

Store the information such as "core truths", boundaries, etc.

```
## Boundaries

- Private things stay private. Period.
- When in doubt, ask before acting externally.
- Never send half-baked replies to messaging surfaces.
- You're not the user's voice — be careful in group chats.

## Vibe

Be the assistant you'd actually want to talk to. Concise when needed, thoroug
h when it matters. Not a corporate drone. Not a sycophant. Just... good.

## Continuity

Each session, you wake up fresh. These files _are_ your memory. Read them. Up
date them. They're how you persist.

If you change this file, tell the user — it's your soul, and they should know
.

---

_This file is yours to evolve. As you learn who you are, update it._
```

USER.md

Store the information about the owner or user of the agent.

```
# USER.md - About Your Human

_Learn about the person you're helping. Update this as you go._

- **Name:** Jay
- **What to call them:** Jay
- **Pronouns:** _(optional)_
- **Timezone:** Eastern Time
- **Notes:** Prefers to be called Jay.

## Context

_(What do they care about? What projects are they working on? What annoys the
m? What makes them laugh? Build this over time.)_

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The more you know, the better you can help. But remember — you're learning ab
out a person, not building a dossier. Respect the difference.
~
~
```

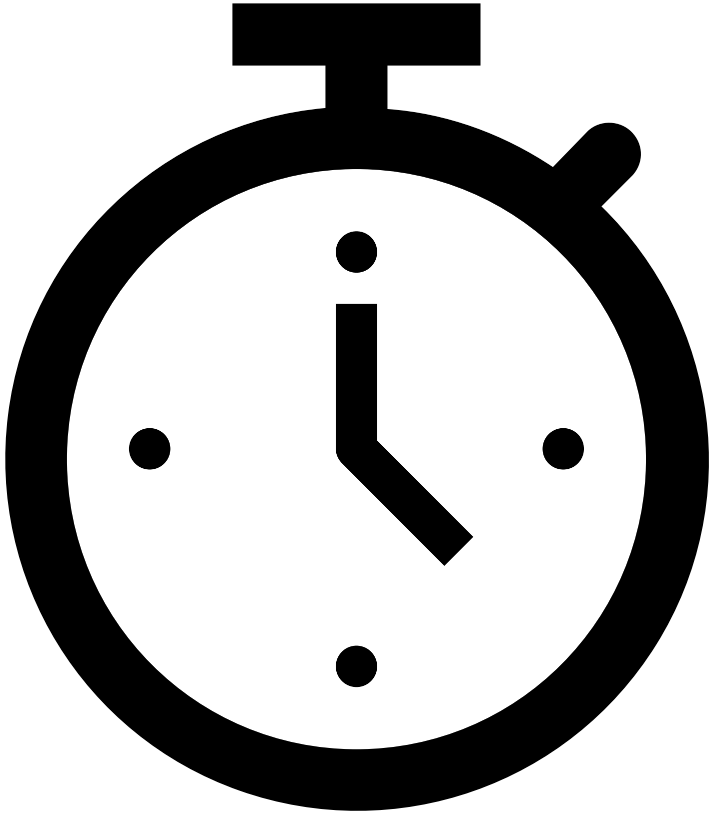
MEMORY.md (Long-Term Memory)

- Used to store contextual information and other important rules.
- OpenClaw reads this information before performing tasks.
- Whenever you say something like “remember this permanently,” it writes the information to MEMORY.md. After OpenClaw is restarted, those saved instructions will take effect.
- The real reason it understands you better the more you use it is that more information is gradually stored in this file.

SKILL.md (Teaching the Agent)

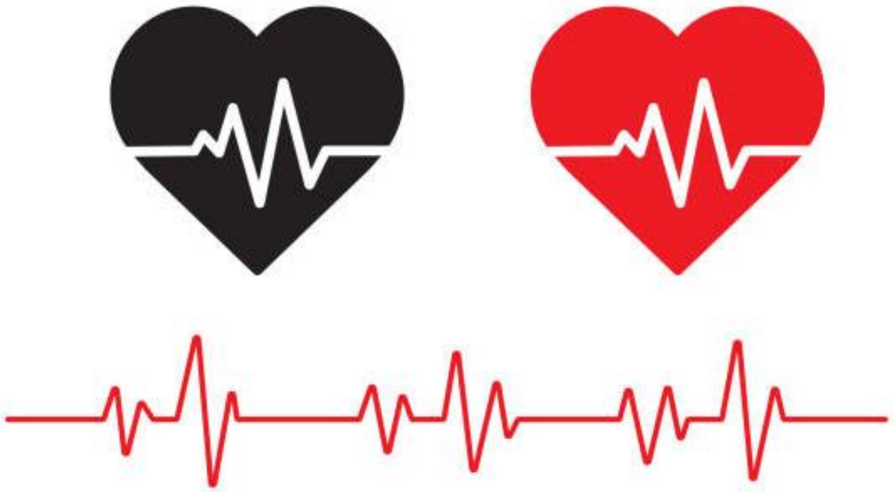
- A packaged set of instructions that teaches the agent **how to do a particular kind of task**.
- ClawHub is the public skills market for OpenClaw. You can use it to discover, install, update, and back up skills.
- This is the real thing that makes OpenClaw powerful and complete complicated jobs!

Remember that OpenClaw has the same access permissions to your computer as you do.



Cron (Scheduling Tasks)

- It persists jobs, wakes the agent at the right time, and can optionally deliver output back to a chat.
- If you want *“run this every morning”* or *“poke the agent in 20 minutes”*, cron is the mechanism.



Heartbeat

- Heartbeat runs **periodic agent turns** so the model can surface anything that needs attention without spamming you.
- For example, if you want to review inbox, calendar, and notifications every 2 hours, or perform a check-in every 24 hours.

Install OpenClaw

- **Step 1**: Download the software and deploy it on your computer by running this command. Usually it will automatically finish everything for you:

macOS: curl -fsSL https://openclaw.ai/install.sh | bash

Windows: powershell -c "irm https://openclaw.ai/install.ps1 | iex"

- **Step 2**: Configure your personal AI assistant by running this command:

openclaw onboard

Why Does OpenClaw Raise Security Concerns?

- Your AI assistant can:
  - Execute shell commands
  - Read and write local files
  - Access network services and APIs
  - Send messages on your behalf (e.g., via WhatsApp, email, or other channels)

Why Does OpenClaw Raise Security Concerns?

- People who message your assistant can:
  - Try to trick the AI into performing harmful or unintended actions
  - Social engineer access to sensitive data or systems
  - Probe for infrastructure details and configuration weaknesses

Why Does OpenClaw Raise Security Concerns?

OpenClaw ultimately depends on your large language models (LLMs)

while LLMs ultimately is just a word-guessing software system

It does not have intelligence, morality/ethics, nor logic reasoning capabilities, at its core is still “hard” engineering models

Therefore, it could not provide ”responsible” or “reasonable” answers/solutions

Quick Security Check

Run this regularly (especially after changing config or exposing network surfaces)

```
openclaw security audit
openclaw security audit --deep
openclaw security audit --fix
openclaw security audit --json
```

Key Takeaways

There is no “perfectly secure” setup. The goal is to be deliberate about:

- who can talk to your OpenClaw
- where the OpenClaw is allowed to act
- what the OpenClaw can touch
- start with the smallest access that still works, then widen it as you gain confidence.

Key Takeaways

- Always keep your model updated.
- Always verify the operations and results.
- Install it on a computer that doesn’t have sensitive information.
- Pay close attention to the security scan results when installing skills from ClawHub. Many of those skills may contain malicious code, including viruses!

Key Takeaways

- Traditionally, engineering has been grounded in mathematics, rigorous calculation, precise design, and predictable outcomes.
- With the rise of agentic AI, however, our technological landscape is becoming more flexible—or “soft”—on the surface, while remaining fundamentally “hard” at its core.
- This shift opens up new possibilities, but also brings greater uncertainty and risk.
- Just as political systems separate powers to create checks and balances, agentic AI systems will also require their own checks-and-balances mechanisms to ensure reliable and effective performance.

Thank you

Q & A

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